Modality Decoupling and Coupling Network for Visible-Infrared Person Re-Identification

Rishen Zhong, Guorong Lin, and Zhenhua Huang

***Abstract*—Visible-Infrared person re-identification (VI-ReID) is a critical cross-modal retrieval task with substantial applications in urban surveillance and public safety. Existing methods primarily introduce text prompts that entangle modality and identity information, failing to clearly separate person uniqueness from modality differences. Additionally, the integration of modality-shared features with modality-specific features ignores the fusion of high-level semantic relationships, leading to the distortion of implicit cross-modal information and weakening identity retrieval. To address these issues, we develop a Modality Decoupling and Coupling Network (MDCN) that aims to fully extract both modality-shared and modality-specific features, while effectively integrating valuable high-level semantic relationships from both. Specifically, it consists of three core components: Context Prompt Decoupling Learning (CPDL), Semantic Coupling Learning (SCL), and Modality Relational Knowledge Distillation (MRKD). CPDL designs modality-contextual text prompts to explicitly introduce modality-aware capabilities and semantically decouples them into person-related and modality-related prompts, thereby effectively distinguishing between modality and identity information. SCL utilizes decoupled semantic prompts as high-level priors while employing modality coupling and modality-contextual alignment to guide visual features. This encourages the model to emphasize modality uniqueness and differences across semantic levels while preserving high-level semantic context, enabling it to learn more stable modality-specific and shared representations. MRKD employs hierarchical cosine angular distillation to project inter-modality and intra-modality semantic relationships into a modality-contextual space, explicitly reducing cross-modal and intra-modal interference while preserving discriminative representations. Extensive experiments on SYSU-MM01, RegDB, and LLCM benchmarks demonstrate that MDCN achieves state-of-the-art performance, validating its effectiveness in addressing the VI-ReID task.**

***Index Terms*—Person re-identification, high-level semantics, relational knowledge distillation.**

I. Introduction

PERSON re-identification (ReID) aims to match pedestrian images of the same individual across non-overlapping camera views. This enhances the operational efficacy of urban surveillance and public safety in applications such as locating missing persons and tracking criminals. Current research efforts [1], [2], [3], [4], [5], [6] have achieved significant progress in the visible spectrum modality. Notably, modern public safety systems extensively use adaptive spectral perception cameras that automatically switch to infrared modality in low-light conditions, effectively preserving pedestrian features at night. However, the significant domain differences between visible and infrared modality images lead to substantial performance degradation when applying conventional ReID frameworks. Therefore, visible-infrared person re-identification (VI-ReID) has emerged as a critical research area to overcome this technical challenge.

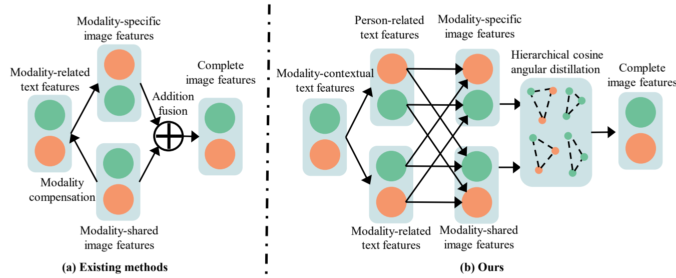

Fig. 1: Illustration of the differences between our method and existing VI-REID methods. Green represents the visible modality, while red represents the infrared modality. (a) Existing methods primarily rely on modality-related text prompts to compensate for modality-specific features and additional fusion to extract complete image features. (b) Our method employs modality-contextual text prompts to semantically decouple person-related and modality-related prompts and use both to guide the extraction of modality-shared and modality-specific features. In addition, Hierarchical Cosine Angular Distillation is introduced to integrate the effective high-level semantic information from both features.

The VI-ReID task aims to effectively alleviate inter-modal representation discrepancies while preserving identity-aware valuable information across modalities [7], [8]. Due to distinct spectral response mechanisms, significant differences exist across modalities in visual features such as color distributions, texture details, etc. The mainstream technologies for solving this problem are divided into Image-level and Feature-level methods [9]. Image-level methods aim to achieve pixel-level cross-modality translation or generate intermediate modalities [10], [11], [12]. Although these methods can effectively alleviate modal discrepancies, their training process is unstable and relies on large-scale cross-modality image pairs. Feature-level methods implement modality-shared representations in deep feature spaces through heterogeneous feature extraction networks and cross-modality metric learning algorithms [13], [14], [15]. However, relying solely on modality-shared features for retrieval ignores identity-aware discriminative information hidden within modality-specific features. This limitation restricts their ability to capture complex identity

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cues. As a result, researchers have started exploring implicit information within modality-specific features and combining it with modality-shared features to create more comprehensive representations [16], [17], [18]. These methods are referred as modality-compensated methods.

With the advancement of Vision-Language Pre-training (VLP) models, researchers have begun incorporating vision-language learning into feature-level methods to compensate for insufficient high-level semantics [19], [20], [21]. In particular, recent studies have shown that Contrastive Language-Image Pre-training (CLIP) models can bridge visual content with linguistic descriptions to enhance the perception of high-level semantics in pedestrians [22], [23], [24]. In this context, modality-compensated methods have been proposed. They introduce high-level semantic information to enhance modality perception, thereby further extracting modality-specific and modality-shared features to obtain more comprehensive pedestrian representations [25]. However, as illustrated in Figure 1 (a), existing methods employ text prompts that primarily focus on modality differences, entangling identity-related and modality information. The direct use of these prompts to guide the model results in modality-shared and modality-specific features losing crucial identity information and introducing noise, ultimately limiting discriminative capability. Additionally, integrating modality-shared features with modality-specific features ignores the fusion of high-level semantic relationships within and across modalities, leading to distortion of cross-modal high-level semantics during information integration, which further weakens the effectiveness of identity retrieval.

To address the above deficiencies, as illustrated in Figure 1 (b), we propose a design idea of modality-contextual text prompts that explicitly introduce modality-aware capabilities. We find that although this enhances the flexibility of multi-modal attribute localization, it also introduces new complexities in maintaining cross-modal representation consistency. To alleviate this dilemma, we develop a Modality Decoupling and Coupling Network (MDCN), which integrates Context Prompt Decoupling Learning (CPDL), Semantic Coupling Learning (SCL), and Modality Relational Knowledge Distillation (MRKD). Specifically, CPDL introduces modality-contextual text prompts to explicitly infuse modality-aware features and semantically decouples them into person-related and modality-related prompts. This decoupling ensures that modality information is effectively distinguished from identity information. SCL employs decoupled semantic prompts as high-level semantic priors, providing guidance for visual feature learning. Modality coupling alignment is designed to encourage the model to emphasize modality uniqueness and differences across semantic levels. Modality-contextual alignment is designed to preserve high-level semantic context. Consequently, the model retains contextual information while learning more stable modality-specific and modality-shared representations. MRKD leverages modality-contextual features with high-level semantic context and employs hierarchical cosine angular distillation to integrate intra- and inter-modality semantic relationships from both modality-specific and modality-shared feature spaces into the modality-contextual space, explicitly reducing cross-modal and intra-modal interference while preserving discriminative representations. Thus, MDCN reduces cross-modal and intra-modal interference, enhances the discriminability of modality-contextual features, and produces rich representations that improve identity retrieval.

The main contributions of this paper are as follows:

1) We develop a novel MDCN method that introduces modality-contextual text prompts for VI-ReID and introduce CPDL to semantically decouple these prompts, effectively distinguishing between modality and identity information.
2) We explore CLIP's potential for modeling modality information in VI-ReID and introduce SCL, which leverages decoupled semantic prompts and employs modality coupling alignment and modality-contextual alignment to construct multiple feature spaces. This emphasizes modality uniqueness and differences while preserving high-level semantic context, resulting in more stable modality-specific and modality-shared representations.
3) We design MRKD, which uses hierarchical cosine angular distillation to integrate intra-modality and inter-modality semantic relationships from both modality-specific and modality-shared features into a modality-contextual space, explicitly enhancing modality-contextual features while preserving discriminative representations.
4) We validate the performance of MDCN on three benchmark datasets, highlighting the superiority of our method and establishing a new technical pathway for the VI-ReID task.

II. Related Work

A. Visble-Infrared Person Re-Identification

Person ReID is a critical pedestrian retrieval task [26]. Its core objective is to extract discriminative pedestrian representations. For instance, ATPR [27] leverages explicit attribute guidance and a dedicated Attribute Embedding mechanism to address intra-class attribute variability; while MP-ReID [28] utilizes appearance attributes to construct both explicit and implicit learnable prompts, generating more comprehensive and discriminative person descriptive features. Despite significant advances in visible-light ReID, these methods show limited generalization for VI-ReID tasks. VI-ReID aims to effectively alleviate inter-modal representation discrepancies while preserving identity-aware valuable information across modalities. Mainstream studies are divided into Image-level and Feature-level methods [9].

Image-level methods reduce modality differences through cross-modal style transfer or intermediate image generation [29], [17], [30]. For example, AlignGAN [10] and Hi-CMD [11] perform bidirectional conversion between visible and infrared images, transforming visible images into infrared and vice versa. MMN [12] introduces a nonlinear intermediate modality generator that bridges the visible-infrared cross-modal gap. In addition, DiVE [31] leverages a pre-trained diffusion model and a novel identity-modality decoupling paradigm to synthesize high-fidelity visible-infrared cross-modal paired data, significantly reducing the modality gap for VI-ReID. However, image quality remains challenging, and the reliability of fine-grained local feature remains problematic due to adversarial training instability and latent space noise. In contrast, our method avoids these issues by eliminating the need for image generation.

Feature-level methods create modality-shared representations in deep feature spaces through heterogeneous feature extraction networks and cross-modality metric learning algorithms [8], [13], [32]. Among them, MRCN [9] employs instance normalization and modality recovery-compensation to decouple modality-relevant features from modality-irrelevant ones. SGIEL [33] uses shape-erased feature learning with human parsing priors to decoupling shape-related and shape-erased features, encouraging the exploration of diverse modality-shared features. Although these methods have made significant progress, they primarily rely on modality-shared features for person re-identification and ignore important identity-relevant information embedded in modality-specific features. This limits the ability to fully exploit the potential of cross-modal retrieval performance. To address these issues, several modality-compensated methods, [16], [17], [18], [34] have been proposed. cm-SSFT [18] proposes a cross-modality shared-specific feature transfer algorithm to address for the lack of specific information. IDKL [16] employs implicit knowledge distillation to extract implicit discriminative knowledge from modality-specific features. D^2RL [34] introduces a Dual-level Discrepancy Reduction Learning framework to generate a unified multi-spectral representation. However, these modality-compensated methods often lack high-level semantic information. On the contrary, our method leverages natural language descriptions to supervise cross-modal semantic representation learning, thereby embedding high-level semantic information into both modality-shared and modality-specific features.

B. Vision-Language Pre-Training

Vision-Language Pre-Training (VLP) models are able to capture high-level semantic information from images by establishing semantic connections between natural language and visual content. However, prompt engineering for downstream tasks faces significant challenges, as it requires domain expertise, is time-consuming to design, and suffers from limited model generalization capability. To address this, Zhou et al. [35] propose CoOp, which utilizes a set of learnable prompts to replace manual text prompts. CoCoOp[36] further generates input-conditional tokens for each image and integrates them into the learnable prompts, enhancing the model's generalization ability. MaPLe[37] introduces a Vision-Language coupling strategy to strengthen the coupling between visual and language prompts. These methods enable VLP models to be directly applied to ReID tasks. For instance, CLIP-ReID[5] generates identity-related language descriptions through learnable prompts and guide the visual encoder to extract semantically rich features. It significantly improves the representation capability for ReID and provides abundant high-level semantic information. However, in VI-ReID, the domain gap between visible and infrared images limits the model's ability to generate detailed image descriptions via text tokens.

Several methods have explored applying VLP to the VI-ReID task [38], [39]. For example, TMAL[19] aligns visible-infrared features through learnable text descriptions and a dual-module approach, combining modality alignment and identity enhancement. Yu et al.[20] find that high-level semantic information in images from different modalities does not show significant modality differences, and they enhance cross-modality feature invariance with a three-stage text feature generation and fusion mechanism. However, these methods rely solely on modality-shared features for person retrieval and ignore the identity-aware discriminative information in modality-specific features, which limits further improvement in retrieval performance. As a result, researchers have combined modality-compensated methods with VLP, such as CLIP-MC[25] uses prompt learning to generate fine-grained text descriptions and recovers missing modality-specific information through Instance Text Prompt Generation. Although existing modality-compensated methods have made progress after incorporating high-level semantic information, they still face two limitations. First, after incorporating high-level semantic information, these methods often fail to effectively separate identity information from modality information. Second, the integration of modality-shared and modality-specific features in these methods generally overlooks the fusion of high-level semantic relationships both within and across modalities. To address these issues, our method fully extracts both modality-shared and modality-specific features, while effectively integrating high-level semantic relationships from both.

III. Method

This section introduces the proposed Modality Decoupling and Coupling Network (MDCN) and details its key components. As shown in Figure 2, a pretrained CLIP [22] is used as the backbone for text feature extraction. MDCN first introduces Context Prompt Decoupling Learning (CPDL), which semantically decouples modality-contextual text prompts into person-related and modality-related prompts and uses modality orthogonality to eliminate cross-modal feature redundancy. This effective decoupling ensures that modality information is clearly distinguished from identity information. Then, MDCN distinguishes modality-specific, modality-shared, and modality-contextual features through three network branches constrained by Semantic Coupling Learning (SCL). High-level semantic information is embedded via modality coupling and modality-contextual alignment and effectively guide the image encoder to focus on key attributes. Meanwhile, common ReID losses are used to boost the discriminative power of the learned features. Finally, MDCN introduces Modality Relational Knowledge Distillation (MRKD), which uses hierarchical cosine angular distillation to distill valuable high-level semantic relationships from modality-specific and modality-shared branches into modality-contextual image features. This enhances the modality-aware distinguishability in modality-contextual representation and ensures a more refined representation of cross-modal information.

Formally, we define the VI-ReID dataset as $\mathcal{D}_I = \{\mathcal{X}_v, \mathcal{X}_r, \mathcal{Y}_v, \mathcal{Y}_r\}$, where $\mathcal{X}_v = \{x_v^i\}_{i=1}^{N_v}$ and $\mathcal{X}_r = \{x_r^i\}_{i=1}^{N_r}$

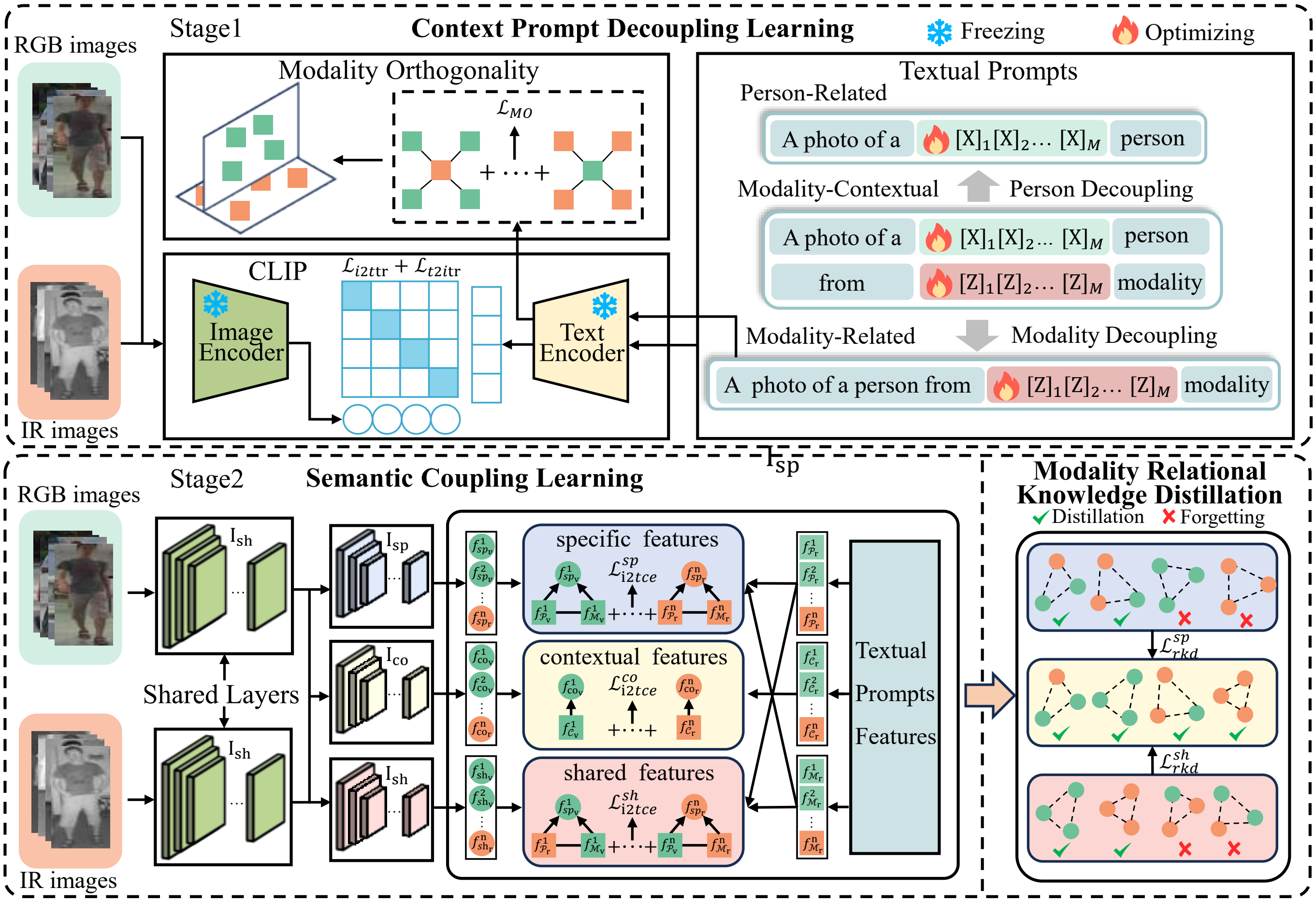


Fig. 2: Overview of the proposed method. The entire learning process of MDCN includes two stages: Stage I (CPDL) and Stage II (SCL+MRKD). CPDL: Given image samples across modalities, we design prompt decoupling learning to generate natural language descriptions corresponding to visible and infrared images respectively. SCL: Considering the semantic coupling of descriptions across different modalities, we introduce modality coupling alignment and modality-contextual alignment to effectively capture and learn their semantic details. MRKD: With the guidance of hierarchical cosine angular distillation, we distills inter-modality cosine relationships from modality-shared features and intra-modality cosine relationships from modality-specific features into the modality context space.

denote the sets of visible and infrared images, respectively. Here, x_v^i and x_r^i represent the i-th visible and infrared samples, while N_v and N_r indicate the corresponding sample counts of the two modalities. The identity label sets are given as $\mathcal{Y}_v = \{y_v^i\}_{i=1}^{N_v}$ and $\mathcal{Y}_r = \{y_r^i\}_{i=1}^{N_r}$, which share a unified identity class space across modalities.

A. Context Prompt Decoupling Learning

Traditional learnable text prompts are designed either for modality differences or for identity descriptions, yet they inevitably entangle the two during optimization. A modality-related prompt absorbs identity cues from paired visual features, while an person-related prompt is forced to accommodate the visual discrepancy across modalities. Consequently, the resulting supervision fails to separate what makes a person unique from what makes an image infrared or visible. To address these issues, CPDL introduces modality-contextual text prompts that semantically bridge person-related and modality-related cues, and then explicitly decouple them. This design prevents the passive entanglement described above, allowing the model to capture spectral distribution differences between visible and infrared modalities while better preserving identity information.

In CPDL, we introduce learnable modality-contextual text prompts with the textual description "A photo of a $[X]_1, [X]_2, \ldots, [X]_m$ person from $[Z]_1, [Z]_2, \ldots, [Z]_m$ modality", where $[X]_i$ and $[Z]_i$ ($i \in [1, m]$) represent trainable word tokens corresponding to the person and modality, respectively. m indicates the number of tokens, with their dimensionality matching the word embedding space. To clearly distinguish modality information from identity information, we semantically decouple a modality-contextual text prompt into two parts: a person-related text prompt, "A photo of a $[X]_1, [X]_2, \ldots, [X]_m$ person", and a modality-related text prompt, "A photo of a person from $[Z]_1, [Z]_2, \ldots, [Z]_m$ modality". The decoupling corresponds to the $[X_i]$ and $[Z_i]$ ($i \in [1, m]$) components in the modality-contextual prompt, ensuring semantic alignment while distinguishing modality information from identity information. These prompts are independently constructed for each identity across different modalities. We define the complete set of learnable prompts as $\mathcal{D}_T = \{\mathcal{P}_v^i, \mathcal{P}_r^i, \mathcal{M}_v^i, \mathcal{M}_r^i, \mathcal{C}_v^i, \mathcal{C}_r^i\}$, where $\mathcal{C}$ denotes the modality-contextual prompt set, and $\mathcal{P}$ and $\mathcal{M}$ represent the decoupled person-related and modality-related prompt sets, respectively. Although person-related information is inherently modality-independent, we keep P_v and P_r separate for the two modalities. If a single unified P were shared, it would be forced to reconcile the large visual discrepancy between visible and infrared appearances, diluting modality-specific identity cues. By preserving separate prompts, complete identity information is retained within each modality. In the subsequent coupling stage, these representations are cross-aligned and converge to a unified, modality-invariant identity representation.

The CLIP framework consists of an image encoder $\mathcal{I}(\cdot)$ and a text encoder $\mathcal{T}(\cdot)$. The image encoder processes both visible and infrared image inputs, enabling the learnable tokens to comprehensively capture identity-discriminative information. We freeze the parameters of $\mathcal{I}(\cdot)$ and $\mathcal{T}(\cdot)$, and optimize only the text tokens $[X]_i$ and $[Z]_i$ ($i \in [1, m]$). Based on the contrastive learning strategy of CLIP-ReID [5], we employ image-text contrastive loss $\mathcal{L}_{i2t}$ and text-image contrastive loss $\mathcal{L}_{t2i}$ as optimization objectives:

$$\mathcal{L}_{i2t}^{i}(V^i, T^i) = -\log \frac{\exp(s(V^i, T^i))}{\sum_{a=1}^{n} \exp(s(V^i, T^a))}, \quad (1)$$

$$\mathcal{L}_{t2i}^{y^i}(T^{y^i}) = -\frac{1}{|P(y^i)|} \sum_{p \in P(y^i)} \log \frac{\exp(s(V^p, T^{y^i}))}{\sum_{a=1}^{n} \exp(s(V^a, T^{y^i}))}. \quad (2)$$

Here, the similarity metric is defined as $s(V^i, T^i) = \langle V^i, T^i \rangle$, B denotes the batch size, and V^i and T^{y^i} are the embedding features produced by $\mathcal{I}(\cdot)$ and $\mathcal{T}(\cdot)$, respectively. $P(y^i) = \{ p \in \{1, \ldots, n\} \mid y^p = y^i \}$ contains the indices of all positive samples sharing the identity y^i, and $|\cdot|$ denotes the cardinality of a set.

We compute triplet image-text contrastive loss $\mathcal{L}_{i2ttri}^{i}$ and text-image contrastive loss $\mathcal{L}_{t2itri}^{i}$ incorporating three types of learnable text tokens:

$$\begin{aligned} \mathcal{L}_{i2ttr}^{i} = {} & \mathcal{L}_{i2t}^{i}(f_{\mathcal{I}_v}^{i}, f_{\mathcal{P}_v}^{i}) + \mathcal{L}_{i2t}^{i}(f_{\mathcal{I}_v}^{i}, f_{\mathcal{M}_v}^{i}) + \mathcal{L}_{i2t}^{i}(f_{\mathcal{I}_v}^{i}, f_{\mathcal{C}_v}^{i}) \\ & + \mathcal{L}_{i2t}^{i}(f_{\mathcal{I}_r}^{i}, f_{\mathcal{P}_r}^{i}) + \mathcal{L}_{i2t}^{i}(f_{\mathcal{I}_r}^{i}, f_{\mathcal{M}_r}^{i}) + \mathcal{L}_{i2t}^{i}(f_{\mathcal{I}_r}^{i}, f_{\mathcal{C}_r}^{i}), \end{aligned} \quad (3)$$

$$\begin{aligned} \mathcal{L}_{t2itr}^{y^i} = {} & \mathcal{L}_{t2i}^{y^i}(f_{\mathcal{P}_v}^{y^i}) + \mathcal{L}_{t2i}^{y^i}(f_{\mathcal{M}_v}^{y^i}) + \mathcal{L}_{t2i}^{y^i}(f_{\mathcal{C}_v}^{y^i}) \\ & + \mathcal{L}_{t2i}^{y^i}(f_{\mathcal{P}_r}^{y^i}) + \mathcal{L}_{t2i}^{y^i}(f_{\mathcal{M}_r}^{y^i}) + \mathcal{L}_{i2t}^{y^i}(f_{\mathcal{C}_r}^{y^i}). \end{aligned} \quad (4)$$

Here, $f_{\mathcal{I}_v}^{i}$ and $f_{\mathcal{I}_r}^{i}$ denote the image embeddings of x_v^i and x_r^i, respectively, while $f_{\mathcal{P}_v}^{i}$, $f_{\mathcal{M}_v}^{i}$, and $f_{\mathcal{C}_v}^{i}$ represent the text embeddings of the learnable prompts $\mathcal{P}_v^i$, $\mathcal{M}_v^i$, and $\mathcal{C}_v^i$, respectively. Similarly, $f_{\mathcal{P}_r}^{i}$, $f_{\mathcal{M}_r}^{i}$, and $f_{\mathcal{C}_r}^{i}$ denote the text embeddings corresponding to the infrared modality.

To establish differentiated feature spaces for text representations in visible and infrared modalities and eliminate intermodal feature redundancy, we impose a modality orthogonality constraint on the decoupled modality-related prompts $\mathcal{M}_v^i$ and $\mathcal{M}_r^i$. This constraint forces the text embeddings of different modalities to maintain maximum separation in the feature space, formulated as:

$$\mathcal{L}_{mo}^{i} = \frac{1}{|P(y^i)|} \sum_{a \in P(y^i)} (f_{\mathcal{M}_v}^{i})^T f_{\mathcal{M}_r}^{a}. \quad (5)$$

Consequently, the overall loss for CPDL is:

$$\mathcal{L}_{cpdl} = \mathcal{L}_{i2ttr} + \mathcal{L}_{t2itr} + \mathcal{L}_{mo}. \quad (6)$$

By minimizing $\mathcal{L}_{cpdl}$, gradients backpropagate through the frozen $\mathcal{I}(\cdot)$, thereby optimizing the learnable prompt set $\mathcal{D}_T$.

B. Semantic Coupling Learning

Effectively embedding high-level semantics into visual representations is challenging in VI-ReID. It is important to emphasize that the integration between textual prompts and visual features in our MDCN is not a trivial feature concatenation, but a two-stage collaborative evolution process with deep geometric mechanisms and practical implications. Specifically, in Stage I (CPDL), the visual features act as a semantic guide to implicitly optimize the continuous text tokens, successfully decoupling entangled identity and modality cues into distinct textual anchors. Building upon these clean anchors, Stage II (SCL) utilizes them as semantically rich, high-level class prototypes. Driven by the prototype-based cross-entropy loss, visual features from both modalities are geometrically constrained to converge toward these frozen textual prototypes. This text-guided integration establishes a domain-invariant coordinate system in the multimodal latent space, strictly regularizing the scattered visual distributions into well-structured semantic clusters rather than passively matching local visual patterns.

Moreover, this text-as-prototype integration mechanism yields significant practical robustness for VI-ReID under real-world surveillance scenarios. Visible and infrared images are highly susceptible to background clutter, severe illumination variations, and sensory noise. Because textual prototypes represent absolute, high-level abstract concepts that are inherently immune to pixel-level distortions, forcing visual features to align with them effectively regularizes the visual encoder to discard irrelevant visual variations. Consequently, even under severe low-light conditions and complex illumination variations, these stable textual prototypes serve as reliable semantic anchors, providing continuous regularization constraints to ensure consistent cross-modal alignment and robust pedestrian retrieval. Guided by this mechanism, SCL restructures the image encoder into three parallel branches, each steered by different prompt pairings to focus on key semantic attributes. By selectively pairing person-related and modality-related prompts, SCL directs each branch to a distinct aspect of the input. The modality-specific features f_{sp} focus on identity cues within each modality, the modality-shared features f_{sh} learn cross-modal invariance, and the modality-contextual features f_{co} integrate both, thereby absorbing complementary relational knowledge from the other two feature spaces and facilitating the learning of implicit cross-modal information. The three branches thus produce complementary, non-redundant representations. During stage II learning, the parameters of the learnable prompt set $\mathcal{D}_T$ and the text encoder $T(\cdot)$ are frozen, and the optimization focuses only on the image encoder. Via the text encoder, SCL derives three types of semantic text representations as guidance. Correspondingly, the output layer of the image encoder is restructured into three parallel branches to extract visual features from both visible and infrared images. The outputs of these branches are:

$$\begin{aligned} f^i_{sp_v} &= \mathcal{I}_{sp}(\mathcal{I}_s(x^v_i)),\ f^i_{sp_r} = \mathcal{I}_{sp}(\mathcal{I}_s(x^r_i)) \\ f^i_{sh_v} &= \mathcal{I}_{sh}(\mathcal{I}_s(x^v_i)),\ f^i_{sh_r} = \mathcal{I}_{sh}(\mathcal{I}_s(x^r_i)), \\ f^i_{co_v} &= \mathcal{I}_{co}(\mathcal{I}_s(x^v_i)),\ f^i_{co_r} = \mathcal{I}_{co}(\mathcal{I}_s(x^r_i)) \end{aligned} \tag{7}$$

where I_s denotes the shared layers (the first three stages of ResNet-50), and I_{sp}, I_{sh}, and I_{co} branch off from Stage 4, corresponding to the modality-specific, modality-shared, and modality-contextual layers, respectively. The image–text alignment cross-entropy loss ξ is used to align visible and infrared modalities with textual information:

$$\xi(f^i_{sp_v}, f^{y^i}_{\mathcal{P}_v}) = \sum_{i=1}^{n_v} q_i \log \frac{\exp\left(s\left(f^i_{sp_v}, f^{y^i}_{\mathcal{P}_v}\right)\right)}{\sum_{y_a=1}^{N_c} \exp\left(s\left(f^i_{sp_v}, f^{y_a}_{\mathcal{P}_v}\right)\right)}, \tag{8}$$

where n_v denotes the number of visible images in a batch. N_c is the total number of identities.

For modality-specific feature learning, we employ modality coupling alignment within each modality, where person-related and modality-related prompts are coupled to compute and match visible and infrared image features. This facilitates intra-modal semantic alignment, as shown below:

$$\begin{aligned} \mathcal{L}^{sp}_{i2tce} = &\ \xi(f_{sp_v}, f_{\mathcal{P}_v}) + \xi(f_{sp_v}, f_{\mathcal{M}_v}) \\ &+ \xi(f_{sp_r}, f_{\mathcal{P}_r}) + \xi(f_{sp_r}, f_{\mathcal{M}_r}). \end{aligned} \tag{9}$$

During modality-shared feature learning, we use modality coupling alignment across modalities, where person-related and modality-related prompts are cross-modally aligned and coupled. This forces the model to capture high-level semantic information across different modalities and facilitates the learning of modality-shared pedestrian representations between visible and infrared images. The loss function is defined as:

$$\begin{aligned} \mathcal{L}^{sh}_{i2tce} = &\ \xi(f_{sh_v}, f_{\mathcal{P}_r}) + \xi(f_{sh_v}, f_{\mathcal{M}_v}) \\ &+ \xi(f_{sh_r}, f_{\mathcal{P}_v}) + \xi(f_{sh_r}, f_{\mathcal{M}_r}). \end{aligned} \tag{10}$$

In addition, since modality-contextual prompts encapsulate richer high-level semantic information, we further perform modality-contextual alignment. It aligns modality-contextual features with modality-contextual prompts to preserve feature integrity and mitigate information loss during decoupling and coupling. Its formula is as follows:

$$\mathcal{L}^{co}_{i2tce} = \xi(f_{co_v}, f_{\mathcal{C}_v}) + \xi(f_{co_r}, f_{\mathcal{C}_r}). \tag{11}$$

The image-text alignment loss and ReID loss are integrated together to obtain the loss functions for each branch:

$$\begin{aligned} \mathcal{L}_{co} &= \mathcal{L}_{reid}(f_{co}) + \mathcal{L}^{co}_{i2tce} \\ \mathcal{L}_{sp} &= \mathcal{L}_{reid}(f_{sp}) + \mathcal{L}^{sp}_{i2tce}, \\ \mathcal{L}_{sh} &= \mathcal{L}_{reid}(f_{sh}) + \mathcal{L}^{sh}_{i2tce} \end{aligned} \tag{12}$$

where $\mathcal{L}_{reid}$ consists of ID loss and weighted triplet loss.

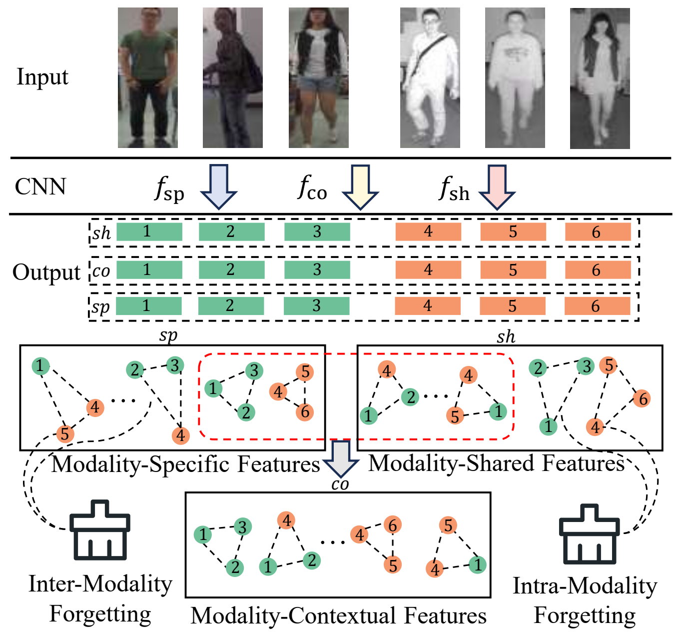


Fig. 3: Illustration of MRKD: Modality-specific features (f_{sp}) emphasize intra-modality discriminability, modality-shared features (f_{sh}) strengthen cross-modal consistency, and modality-contextual features (f_{co}) integrate higher-level semantic cues. MRKD leverages inter- and intra-modality to distill high-level semantic relationships across these spaces.

Finally, the overall loss of the SCL module is:

$$\mathcal{L}_{scl} = \lambda_1 \mathcal{L}_{co} + \lambda_2 \mathcal{L}_{sp} + \lambda_3 \mathcal{L}_{sh}, \tag{13}$$

where λ_1, λ_2, and λ_3 are hyperparameters used to balance each loss term.

C. Modality Relational Knowledge Distillation

SCL produces three complementary feature spaces through prompt coupling and cosine alignment, refining the spatial structure of visual features and enhancing semantic-aware discriminability across all three feature types. However, no single feature space alone suffices for robust retrieval, since each implicitly captures valuable high-level semantic relationships both within and across modalities. As shown in Figure 3, the modality-specific space exhibits compact intra-modal distributions with small angular variations, while nearly orthogonal structures appear across modalities. In contrast, the modality-shared space reveals stable cross-modal correlations but relatively weaker intra-modal discriminability. To exploit these complementary cues, we use the modality-contextual features f_{co} to absorb such implicit knowledge from the other two spaces, equipping it with both intra-modal compactness and cross-modal consistency. f_{co} is supervised by the full modality-contextual prompt C, which fuses person-related and modality-related semantics, making it a holistic representation naturally suited for integrating diverse relational knowledge without introducing redundancy.

To extract and transfer this implicit relational knowledge, we introduce MRKD based on Relational Knowledge Distillation (RKD) [40]. We employs hierarchical cosine angular distillation, which encodes angular relationships from different feature spaces into transferable knowledge representations.

For a given mini-batch, we construct triplet samples (f^i, f^j, f^k), where f^j serves as an anchor reference point. The positive or negative relationship is defined according to whether the samples come from the same modality. Specifically, a pair is considered positive if both samples belong to the same modality (both visible or both infrared), and negative if they belong to different modalities (one visible and one infrared). This modality-based definition allows us to explicitly capture intra-modality consistency and inter-modality discrepancy.

We compute the cosine values of the angles formed in the feature representation space:

$$\begin{aligned} &\cos\angle f^i f^j f^k = \langle e^{ij}, e^{kj} \rangle \\ &e^{ij} = \frac{f^i - f^j}{\|f^i - f^j\|_2},\ e^{kj} = \frac{f^k - f^j}{\|f^k - f^j\|_2}, \end{aligned} \tag{14}$$

where $\langle \cdot, \cdot \rangle$ denotes the vector inner product. The unit vectors e^{ij} and e^{kj} represent the pure directional variations from the reference f^j to f^i and f^k, respectively.

In this geometric formulation, the angle $\angle f^i f^j f^k$ encodes the modality relationship among the three samples. For example, when f^i and f^j are from the same modality (positive pair) and f^k is from the opposite modality (negative pair) relative to f^j, the angle measures whether the intra-modality feature direction is structurally decoupled from the inter-modality feature direction. A near-orthogonal angle indicates that intra-modality variations and inter-modality discrepancies lie in independent directions, reflecting strong modality discriminability. Conversely, when all three samples come from the same modality (all positive), the angle delineates the compactness of intra-modality feature distributions; when they come from three different modality combinations (mixed), the angle reveals the cross-modality distribution structure.

For different feature spaces, we employ a hierarchical angular distillation strategy. In the modality-specific feature space, we extract the angular relationships within each modality and enforce inter-modality forgetting, deliberately discarding cross-modal angular relationships during distillation. This prevents modality-specific features from being contaminated by cross-modal inconsistencies and preserves pure intra-modal discriminative cues. Accordingly, intra-modal angular relation pairs are defined as follows:

$$\begin{aligned} \mathcal{R}_{sp_v}^{ijk} &= [\cos\angle f_{sp_v}^i f_{sp_v}^j f_{sp_v}^k, \cos\angle f_{co_v}^i f_{co_v}^j f_{co_v}^k] \\ \mathcal{R}_{sp_r}^{ijk} &= [\cos\angle f_{sp_r}^i f_{sp_r}^j f_{sp_r}^k, \cos\angle f_{co_r}^i f_{co_r}^j f_{co_r}^k] \end{aligned}. \tag{15}$$

The relational knowledge distillation loss for the modality-specific feature space is:

$$\mathcal{L}_{rkd}^{sp} = \sum_{(i,j,k)\in B} l_\delta\left(\mathcal{R}_{sp_v}^{ijk}\right) + \sum_{(i,j,k)\in B} l_\delta\left(\mathcal{R}_{sp_r}^{ijk}\right), \tag{16}$$

where l_δ denotes the Huber loss function.

Conversely, for the modality-shared feature space, we enforce intra-modality forgetting, deliberately discarding intra-modal angular relationships during distillation. This prevents modality-shared features from being influenced by modality-specific cues and preserves pure cross-modal consistency. The cross-modal angular relation distillation pairs are formulated as:

$$\begin{aligned} \mathcal{R}_{sh_v}^{ijk} &= [\cos\angle f_{sh_r}^i f_{sh_v}^j f_{sh_r}^k, \cos\angle f_{co_r}^i f_{co_v}^j f_{co_r}^k] \\ \mathcal{R}_{sh_r}^{ijk} &= [\cos\angle f_{sh_v}^i f_{sh_r}^j f_{sh_v}^k, \cos\angle f_{co_v}^i f_{co_r}^j f_{co_v}^k] \end{aligned}. \tag{17}$$

The relational knowledge distillation loss for the modality-shared feature space is:

$$\mathcal{L}_{rkd}^{sh} = \sum_{(i,j,k)\in B} l_\delta\left(\mathcal{R}_{sh}^{v}\right) + \sum_{(i,j,k)\in B} l_\delta\left(\mathcal{R}_{sh}^{r}\right). \tag{18}$$

By combining the distillation objectives of modality-specific and modality-shared feature spaces, the complete loss in MRKD is:

$$\mathcal{L}_{mrkd} = \mathcal{L}_{rkd}^{sp} + \mathcal{L}_{rkd}^{sh}. \tag{19}$$

D. Training and Inference

The overall training pipeline of MDCN is outlined in Algorithm 1. In the first stage, CPDL initializes image and text encoders via the pre-trained CLIP and introduces a set of learnable text prompts, including person-related, modality-related, and modality-contextual text tokens. These prompts are iteratively optimized based on the corresponding image features using the objectives defined in Eqs. (6), thus facilitating the acquisition of semantic representations. Based on this, the second stage introduces SCL and MRKD to further refine the feature space. SCL uses modality coupling alignment and modality-contextual alignment to optimize three feature spaces, as defined in Eq.(13). MRKD adopts a hierarchical cosine angular distillation strategy to distill angular correlations among triplet samples into transferable relational knowledge, thereby facilitating the joint optimization of valuable high-level semantic relationships in both modality-shared and modality-contextual representations, as described in Eq.(19).

During inference, we exclusively utilize the modality-contextual features (f_{co}) extracted by the image encoder for cross-modality person matching. To enhance the robustness of metric learning, we introduce a hybrid distance metric that combines Euclidean distance with cosine similarity:

$$D_{\text{joint}} = 0.3 \cdot \Delta\left(f_{co}^q, f_{co}^g\right) + 0.7 \cdot \Theta\left(f_{co}^q, f_{co}^g\right). \tag{20}$$

Here $\Delta(\cdot, \cdot)$ denotes the Euclidean distance and $\Theta(\cdot, \cdot)$ represents the cosine distance. f_{co}^q and f_{co}^g denote the query and gallery modality-contextual features, respectively.

Euclidean distance preserves the absolute magnitude relationships of features, while cosine distance emphasizes angular similarity in the embedding space. By balancing these two complementary metrics, we can effectively enhance both discriminability and stability in cross-modal retrieval.

IV. Experiments

A. Datasets and Experimental Settings

Datasets. We conduct experiments on three benchmark datasets: SYSU-MM01 [41], RegDB [55], and LLCM [51].

TABLE I: Performance comparison with state-of-the-art methods on SYSU-MM01: CMC (%) and mAP (%).

Methods	All-search								Indoor-search							
	Single-shot				Multi-shot				Single-shot				Multi-shot			
	r=1	r=10	r=20	map	r=1	r=10	r=20	map	r=1	r=10	r=20	map	r=1	r=10	r=20	map
Zero-Padding [41]	14.80	54.12	71.33	15.95	19.13	61.40	78.41	10.89	20.58	68.38	85.79	26.92	24.43	75.86	91.32	18.86
D-HSME [42]	20.68	62.74	77.95	23.12	-	-	-	-	-	-	-	-	-	-	-	-
AlignGAN [10]	42.40	85.00	93.70	40.70	51.50	89.40	95.70	33.90	45.90	87.60	94.40	54.30	57.10	92.70	97.40	45.30
DDAG [43]	54.75	90.39	95.81	53.02	-	-	-	-	61.02	94.06	98.41	67.98	-	-	-	-
NFS [44]	56.91	91.34	96.52	55.45	63.51	94.42	97.81	48.56	62.79	96.53	99.07	69.79	70.03	97.70	99.51	61.45
MCLNet [45]	65.40	93.33	97.14	61.98	-	-	-	-	72.56	96.98	99.20	76.58	-	-	-	-
CAJ [46]	69.88	95.71	98.4	66.89	-	-	-	-	76.26	97.88	99.49	80.37	-	-	-	-
MPANet [47]	70.58	96.21	98.80	68.24	75.58	97.91	99.43	62.91	76.74	98.21	99.57	80.95	84.22	99.66	99.96	75.11
FMCNet [17]	66.34	-	-	62.51	73.44	-	-	56.06	68.15	-	-	74.09	78.86	-	-	63.82
CMT [48]	71.88	96.45	98.87	68.57	80.23	97.91	99.53	63.13	76.9	97.68	99.64	79.91	84.87	99.41	99.97	74.11
DSCNet [49]	73.89	96.27	98.84	69.47	-	-	-	-	79.35	98.32	99.77	82.65	-	-	-	-
MSCLNet [50]	76.99	97.63	99.18	71.64	-	-	-	-	78.49	**99.32**	**99.91**	81.17	-	-	-	-
DEEN [51]	74.7	97.6	99.2	71.8	-	-	-	-	80.3	99.0	99.8	83.3	-	-	-	-
SAAI [52]	75.90	-	-	77.03	82.8	-	-	82.39	83.20	-	-	88.01	90.73	-	-	**91.30**
SGIEL [33]	77.12	97.03	99.08	72.33	-	-	-	-	82.07	97.42	98.87	82.95	-	-	-	-
PartMix [13]	77.78	-	-	74.62	80.54	-	-	69.84	81.52	-	-	84.38	87.99	-	-	79.95
DMA [7]	74.57	-	-	70.41	83.03	-	-	65.55	82.85	-	-	85.10	91.33	-	-	80.49
WRIM-Net [53]	77.40	-	-	75.40	83.20	-	-	71.10	86.20	-	-	88.10	92.10	-	-	85.60
IDKL [16]	81.42	97.38	98.89	79.85	84.34	98.89	99.73	78.22	87.14	98.28	99.26	89.37	94.30	99.71	99.93	88.75
DSAF [54]	76.42	97.71	-	73.24	83.48	**99.27**	-	**83.78**	**92.62**	98.20	-	86.37	93.25	98.98	-	87.98
DiVE [31]	79.07	-	-	74.96	-	-	-	-	82.98	-	-	85.90	-	-	-	-
MIP [39]	70.84	-	-	66.41	74.68	-	-	58.49	78.80	-	-	79.92	82.29	-	-	70.65
TMAL [19]	70.58	96.67	98.93	73.96	-	-	-	-	76.17	98.56	99.79	82.70	-	-	-	-
CSDN [20]	75.2	96.6	98.8	71.8	80.6	98.3	99.7	66.3	82.0	98.7	99.5	85.0	88.5	99.6	99.9	80.4
CLIP-MC [25]	78.06	**97.93**	99.32	76.49	-	-	-	-	84.51	98.75	99.82	87.02	-	-	-	-
MDCN (Ours)	**82.67**	97.78	**99.41**	**80.24**	**85.17**	98.92	**99.78**	79.56	88.04	98.79	99.55	**89.83**	**94.83**	**99.86**	**99.97**	88.95

Algorithm 1 MDCN's Training Process

Input: Training image sample set $\mathcal{D}_I$, text encoder $\mathcal{T}(\cdot)$ and image encoder $\mathcal{I}(\cdot)$ from Pre-trained CLIP.
Output: Trained visual encoders $\mathcal{I}_{sp}$, $\mathcal{I}_{sh}$, $\mathcal{I}_{co}$ and $\mathcal{I}_s$.

Stage I: Training CPDL
Initialize learnable text prompts $\mathcal{D}_T$
Freeze parameters in $\mathcal{T}(\cdot)$ and $\mathcal{I}(\cdot)$
while Training is not over **do**
  Extract image features $f^i_{\mathcal{I}} = \mathcal{I}(x^i)$
  Obtain modality-contextual text features $f^i_{\mathcal{C}} = \mathcal{T}(\mathcal{C}^i)$, modality-contextual text features $f^i_{\mathcal{P}} = \mathcal{T}(\mathcal{P}^i)$ and modality-shared text features $f^i_{\mathcal{M}} = \mathcal{T}(\mathcal{M}^i)$
  Compute $\mathcal{L}_{i2ttr}$, $\mathcal{L}_{t2itr}$ and $\mathcal{L}_{mo}$ to update learnable text tokens set by minimizing Ep.(6)
end while
Stage II: Training SCL and MRKD
Freeze parameters in $\mathcal{T}(\cdot)$ and learnable text tokens in $\mathcal{D}_T$
Extract text features $f_{\mathcal{C}}$, $f_{\mathcal{P}}$ and $f_{\mathcal{M}}$ for all IDs
while Training is not over **do**
  Extract modality-shared features $f^i_{sh} = \mathcal{I}_{sh}(\mathcal{I}_{sh}(x^i))$, modality-specific features $f^i_{sp} = \mathcal{I}_{sp}(\mathcal{I}_{sh}(x^i))$ and modality-contextual features $f^i_{co} = \mathcal{I}_{co}(\mathcal{I}_{sh}(x^i))$
  Compute $\mathcal{L}_{co}$, $\mathcal{L}_{sp}$ and $\mathcal{L}_{sh}$ to update f^i_{sh}, f^i_{sp} and f^i_{co} by minimizing Ep.(13)
  Optimize modality-contextual features f^i_{co} by obtaining semantic-aware discriminability from f^i_{sp} and f^i_{sh} by minimizing Ep.(19)
  Optimize visual encoders $\mathcal{I}_{co}$, $\mathcal{I}_{sp}$, $\mathcal{I}_{sh}$ and $\mathcal{I}_s$
end while

SYSU-MM01, the most challenging large-scale benchmark for VI-ReID, contains 287,628 visible images and 15,792 infrared images from 491 identities. It is captured through 4 visible cameras and 2 infrared cameras in indoor/outdoor hybrid environments. The training set contains 22,258 visible images and 11,909 infrared images from 395 identities. The query set includes 96 identities with 3,803 infrared images, while the gallery set consists of 301 and 3,010 randomly selected visible images. The evaluation protocol employs a reversed cross-modality retrieval setting where infrared images query visible images. Two test scenarios are provided: all-search (indoor/outdoor) and indoor-search (indoor cameras only). For gallery construction, single-shot (one image per identity) and multi-shot (ten images per identity) sampling strategies are supported. We comprehensively evaluate performance under all these configurations.

RegDB, a small-scale VI-ReID dataset, is collected through one visible camera and one paired infrared camera. It contains 8,240 images of 412 identities, with 206 identities used for training and 206 identities for testing. Each identity has 10 visible and 10 infrared images. The training set consists of images from 206 randomly selected identities, and the test set is formed from the images of the other 206 identities. Two test modes are evaluated, consisting of visible-to-infrared retrieval and infrared-to-visible retrieval. For both modes, we perform 10 test iterations and report the average results to ensure accuracy and fairness.

LLCM, the low-light cross-modality dataset, is collected under extremely low-light conditions through 9 visible/infrared cameras. It contains 46,767 images of 1,064 identities, with each identity captured by at least one visible and one infrared camera. The training set has 30,921 images from 713 identities, including 16,946 visible images and 13,975 infrared images; while the test set contains 15,846 images from 351 identities, with 8,680 visible images and 7,166 infrared images. Both visible-to-infrared and infrared-to-visible modes are used to evaluate the performance of VI-ReID models. This process is repeated 10 times with different random gallery splits, and the average results are reported to ensure reliability and fairness.

Evaluation metrics. Following previous works, Cumulative Matching Characteristics (CMC) and mean Average Precision (mAP) are used as evaluation metrics in this paper. CMC-k (i.e., Rank-k) calculates the probability that a correct match appears in the top-k retrieval results, while mAP offers a comprehensive evaluation of retrieval performance.

TABLE II: Performance comparison with state-of-the-art methods on RegDB: CMC (%) and mAP (%).

Methods	Visible to Infrared		Infrared to Visible	
	Rank-1	mAP	Rank-1	mAP
Zero-Padding [41]	17.8	18.9	16.7	17.9
AlignGAN [10]	57.9	53.6	56.3	53.4
DDAG [43]	69.34	63.46	68.06	61.80
MCLNet [45]	80.31	73.07	75.93	69.49
MPANet [47]	83.7	80.9	82.8	80.7
SMCL [56]	83.93	79.83	83.05	78.57
CAJ [46]	85.03	77.82	84.75	77.82
MSCLNet [50]	84.17	80.99	83.86	78.31
MID [57]	87.45	84.85	84.29	81.41
FMCNet [17]	89.12	84.43	88.38	83.86
CMT [48]	95.17	87.3	91.97	84.46
SGIEL [33]	91.07	85.23	92.18	86.59
SAAI [52]	91.07	**91.45**	92.09	**92.01**
DEEN [51]	91.1	85.1	89.5	83.4
DARD [58]	86.19	85.39	85.53	85.09
WGCN [8]	90.61	84.53	88.77	81.61
DMA [7]	93.30	88.34	91.50	86.80
WRIM-Net [53]	94.5	90.5	93.7	89.7
IDKL [16]	94.72	90.19	94.22	90.43
DSAF [54]	92.62	86.37	93.25	87.17
CM-DASN [59]	94.45	91.39	93.66	91.41
MIP [39]	92.38	85.99	91.26	85.90
TMAL [19]	82.65	85.04	80.05	78.57
CSDN [20]	89.0	84.7	88.2	82.8
CLIP-MC [25]	80.02	77.23	80.37	76.62
MDCN (Ours)	**95.13**	90.78	**94.88**	90.57

Implementation Details. Our proposed method and all experiments are implemented on a single NVIDIA GeForce 4090 GPU using the PyTorch framework. The network adopts a three-stream architecture, with a modified ResNet-50 pre-trained on CLIP serving as the backbone for feature extraction. Input images are resized to 384×144, and data augmentation strategies such as random flipping, grayscaling [60], and padding are applied to enrich the diversity of training samples. The first training stage runs for 120 epochs, where learnable tokens $[X]_1, [X]_2, \dots, [X]_m$ and $[Z]_1, [Z]_2, \dots, [Z]_m$ $(m = 8)$ are optimized using Adam optimizer [61]. The initial learning rate is set to 3.5×10^{-4} and decays according to the cosine annealing schedule [5]. In the second training stage, only the visual encoder and classifier are trained for 90 epochs, also using Adam optimizer with the same initial learning rate of 3.5×10^{-4}. A warm-up strategy is applied for the first 10 epochs, and the learning rate is reduced by a factor of 10 at the 40th and 70th epochs. For all experiments, the batch size is set to 64, with each batch containing 8 randomly selected identities. Each identity has 4 visible images and 4 infrared images. The hyperparameters are set as $\lambda_1 = 0.7$, $\lambda_2 = 0.5$ and $\lambda_3 = 0.8$. During testing, only modality-contextual features are used for performance evaluation.

TABLE III: Performance comparison with state-of-the-art methods on LLCM: CMC (%) and mAP (%).

Methods	Visible to Infrared		Infrared to Visible	
	Rank-1	mAP	Rank-1	mAP
CAJ [46]	56.5	59.8	48.8	56.6
MMN [46]	59.9	62.7	52.5	58.9
DEEN [51]	62.5	65.8	54.9	62.9
IDKL [16]	**72.22**	66.43	**70.72**	65.19
DSAF [54]	56.38	68.15	56.34	64.27
AGPI^2 [62]	61.78	65.08	56.47	62.79
IRM [63]	66.7	67.5	65.7	67.2
MDCN (Ours)	67.20	**68.75**	66.68	**67.82**

B. Comparison with State-of-the-art Methods

We compare the proposed MDCN with state-of-the-art VI-ReID methods on three VI-ReID datasets. The comparison results on SYSU-MM01, RegDB, and LLCM datasets are reported in Tables I, II, and III, respectively. The competitors include both methods that use learnable text prompts and those that do not. For clarity, the prompt-based methods are listed in the lower half of Tables I and II. The best results are highlighted in bold.

1) Comparison on SYSU-MM01: The experimental results in Table I clearly demonstrate the superior performance of the proposed MDCN over state-of-the-art methods. Under the challenging single-shot all-search setting, MDCN achieves a Rank-1 accuracy of 82.67% and an mAP of 80.24%, outperforming all competitors. In the indoor-search scenario with the same single-shot protocol, the performance of MDCN is further improved, reaching 88.04% in Rank-1 accuracy and 89.83% in mAP. Similarly, under the multi-shot indoor-search protocol, MDCN consistently outperforms the competitors. Notably, MDCN exhibits significant advantages over other prompt-based methods, such as TMAL [19], CSDN [20], and CLIP-MC [25], demonstrating the effectiveness of our method.

2) Comparison on RegDB: As shown in Table II, the proposed MDCN demonstrates outstanding performance on the small-scale RegDB dataset, achieving state-of-the-art results in both retrieval directions. Specifically, in the visible-to-infrared setting, MDCN achieves a Rank-1 accuracy of 95.13%, while in the infrared-to-visible setting, it reaches 94.88%. These results significantly outperform recent leading VI-ReID methods, showcasing the robustness and generalization capability of our method, even in resource-constrained scenarios. MDCN also demonstrates clear advantages over other prompt-based methods, such as MIP [39] and CSDN [20]. This indicates that our proposed representation learning strategy cannot only improve cross-modal feature alignment but also effectively exploit semantic prompts without overfitting on small-scale data.

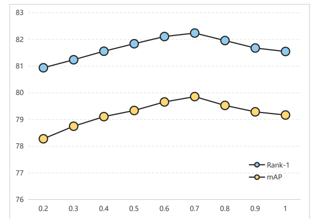
3) Comparison on LLCM: As shown in Table III, a comprehensive evaluation on LLCM dataset confirms the robust generalization capability of MDCN under complex low-light conditions. It achieves mAP values of 68.75% and 67.82% in two retrieval modes, respectively, setting a new state-of-the-art benchmark. This consistent superiority across diverse environmental conditions highlights the effectiveness of our method in handling challenging multi-modal data distributions and illumination variations.

TABLE IV: Ablation experimental results of each component in MDCN on SYSU-MM01.

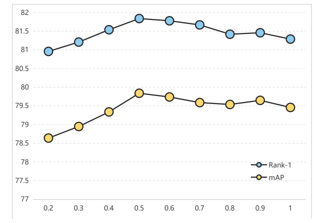
Methods	Base	CPDL		SCL	MRKD	SYSU-MM01	
		$\mathcal{L}_{i2ttr} + \mathcal{L}_{t2itr}$	$\mathcal{L}_{mo}$			Rank-1	mAP
1	✓					73.90	71.40
2	✓	✓		✓		79.27	75.78
3	✓	✓	✓	✓		80.83	77.21
4	✓	✓	✓	✓	✓	**82.67**	**80.24**

TABLE V: Ablation experimental results on the effects of inter/intra-modality constraints in modality-shared and modality-specific features.

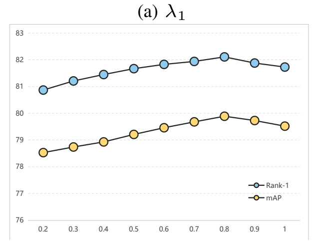
Methods	Modality-Shared		Modality-Specific		SYSU-MM01	
	Inter	Intra	Inter	Intra	Rank-1	mAP
1					80.83	77.21
2			✓	✓	80.58	76.80
3			✓		80.21	76.52
4				✓	81.19	77.51
5	✓	✓			82.18	79.73
6	✓				82.36	79.85
7		✓		✓	81.03	77.37
8	✓		✓		82.12	79.59
9	✓	✓	✓	✓	82.05	79.43
10	✓			✓	**82.67**	**80.24**



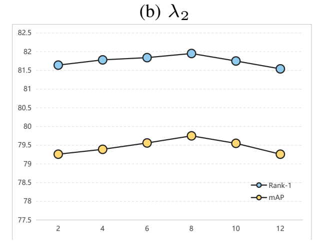
(a) λ_1



(b) λ_2



(c) λ_3



(d) m

Fig. 4: Hyperparameter analysis of MDCN on SYSU-MM01. The figure illustrates the effects of hyperparameters (a) λ_1, (b) λ_2, (c) λ_3, and (d) m on model performance, measured in terms of Rank-1 accuracy and mAP.

C. Ablation Studies

We conduct ablation studies on SYSU-MM01 dataset under the all-search single-shot setting to comprehensively evaluate the contribution of each component in MDCN. The overall experimental configuration remains consistent, and only the evaluated module is modified.

1) Effectiveness of Key Components: We adopt CLIP-VIReID [20] as the baseline model and progressively incorporate each component to evaluate its individual contributions to the performance of MDCN. As shown in Table IV, the inclusion of CPDL (w/o $\mathcal{L}_{MO}$) and SCL significantly improves the performance, with a +5.37% increase in Rank-1 accuracy and a +4.38% gain in mAP compared to the baseline. Further integrating the modal orthogonality constraint into CPDL can improve Rank-1 by an additional +1.56% and mAP by an additional +1.43%. These results validate the importance of explicitly decoupling modality-contextual semantics into person-related and modality-related semantics. Moreover, the results demonstrate that SCL and CPDL work synergistically to fully exploit decoupled semantics, thereby enhancing retrieval robustness. MRKD achieves the most significant improvement in mAP, with a +3.03% increase, highlighting its strong ability to refine the feature space structure. By distilling angular relations from hybrid features, MRKD effectively transfers intra-modality compactness (from modality-specific features) and inter-modality consistency (from modality-shared features) into the modality-contextual feature space. This structured knowledge transfer significantly enhances the discriminative ability on hard samples. Ultimately, MDCN achieves 82.67% in Rank-1 and 80.24% in mAP.

2) MRKD Distillation Targets: We conduct experiments to verify the effectiveness of the MRKD design, which applies intra-modality distillation to modality-specific features and inter-modality distillation to modality-shared features. Table V reports the results. MRKD distills high-level semantic relationships from both modality-shared and modality-specific features into the modality-contextual feature space. The inter-modality constraint enforces spatial relationship alignment across modalities, while the intra-modality constraint preserves consistent angular relationships within each modality. When no relational constraint is applied, the baseline achieves 80.83% Rank-1 and 77.2% mAP. For Methods 2-4, where no constraints are applied to modality-specific features or only the inter-modality constraint is applied, performance slightly decreases. However, when the intra-modality constraint is introduced, Rank-1 accuracy increases by +0.36% and mAP improves by +0.3%. This confirms that modality-specific features contain identity-aware discriminative information, and properly structured constraints are essential for preserving discriminative intra-modal characteristics. Similarly, for Methods 5-6, the results show that modality-shared features contain a significant amount of identity-aware discriminative information. Furthermore, as demonstrated in Methods 7-8, when both inter-modality and intra-modality constraints are applied, Rank-1 accuracy improves by +1.29% and mAP increases by +2.38%. In contrast, the improvement brought by applying the intra-modality constraint alone is small. This suggests that inter-modality constraints play a more substantial role in enhancing the model's ability to align cross-modal features, while intra-modality constraints help preserve fine-grained discriminative details within each modality. Finally, as shown in Methods 9–10, the results demonstrate that our method effectively integrates the semantic-aware discriminability of both feature types, leading to superior cross-modal matching performance.

3) Prompt Design: The ablation study in Table VI investigates the impact of modality-contextual text prompts on the performance of the proposed MDCN. We test different prompt templates to evaluate how the integration of modality-aware semantics influences cross-modal retrieval. Template 1 does not contain modality-related information and therefore achieves only moderate performance. Similarly, Template 2, which introduces additional contextual words such as condition, shows a slight improvement in performance, suggesting that more complex prompt descriptions can help improve overall results. In contrast, when modality-aware prompts are

TABLE VI: Ablation experimental results on the impact of modality-contextual text prompt template design.

No.	Modality-Contextual Text Prompt Template	SYSU-MM01	
		Rank-1	mAP
1	"A $[Z]_1, [Z]_2, \ldots, [Z]_m$ photo of a $[X]_1, [X]_2, \ldots, [X]_m$ person"	81.78	79.50
2	"A photo of a $[X]_1, [X]_2, \ldots, [X]_m$ person obsered in $[Z]_1, [Z]_2, \ldots, [Z]_m$ condition"	81.86	79.61
2	"$[X]_1, [X]_2, \ldots, [X]_m$ person, $[Z]_1, [Z]_2, \ldots, [Z]_m$ modality"	82.13	79.86
3	"A photo of a $[X]_1, [X]_2, \ldots, [X]_m$ person from $[Z]_1, [Z]_2, \ldots, [Z]_m$ modality"	**82.67**	**80.24**

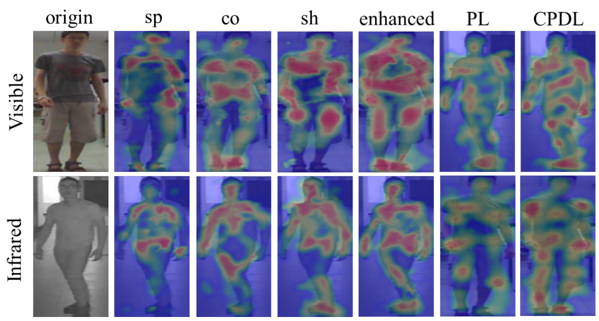


Fig. 5: Visualization of feature activation maps in MDCN for both visible and infrared modalities. From left to right: original input image, modality-specific (sp), modality-contextual (co), modality-shape (sh), enhanced modality-contextual (enhanced), prompt learning (PL), and context prompt decoupling learning (CPDL).

explicitly incorporated, as shown in Templates 3 and 4, the model achieves a clear performance boost. Notably, Template 4 outperforms Template 2, with an improvement of +0.81% in Rank-1 accuracy and +0.63% in mAP. These findings confirm the critical role of modality-aware prompts in enhancing cross-modal matching, as they guide the model to capture modality-specific cues more effectively, thereby strengthening feature alignment across heterogeneous modalities.

D. Hyperparameter Analysis

We perform a detailed sensitivity analysis to evaluate the effects of different hyperparameters on the proposed SCL. In particular, we investigate the impact of different loss weights assigned to modality-specific, modality-shared, and modality-contextual features, as well as the impact of the number of tokens used in text prompts. This analysis provides a deeper understanding of how different hyperparameter settings affect overall performance and helps determine a balanced configuration for achieving optimal results.

1) Effect of λ_1: As shown in Figure 4 (a), the performance steadily increases as λ_1 grows from 0.2 to 0.7, peaking at a Rank-1 accuracy of 82.24% and an mAP of 79.86%. Beyond this point, further enlarging λ_1 causes a slight decline in performance, indicating that excessive emphasis on contextual information may suppress other semantic cues. These results suggest that adopting a moderate value of λ_1 is essential for maintaining a balanced representation across modalities.

2) Effect of λ_2: From Figure 4 (b), we observe a similar trend in which performance improves as λ_2 increases, reaching a peak around 0.5 with a Rank-1 accuracy of 81.84% and an mAP of 79.84%. When the weight continues to grow, performance declines slightly, suggesting that overly strong modality-specific constraints may hinder the integration of shared semantic features. This analysis highlights the importance of assigning a moderate weight to modality-specific features to enhance discriminability while preserving generalization capability.

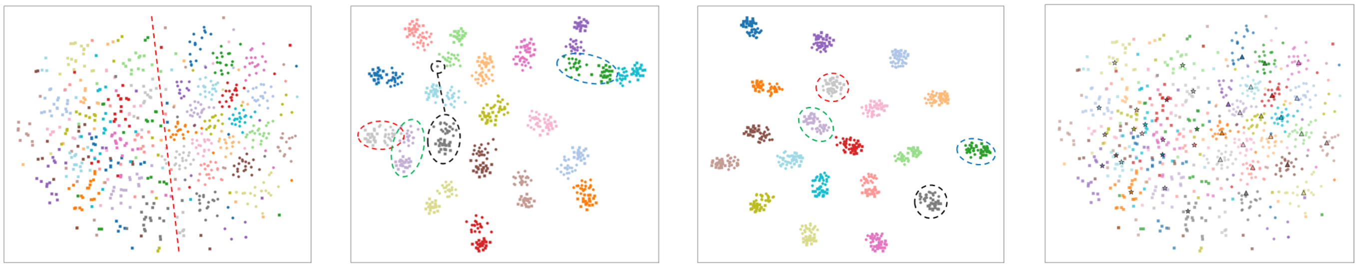
3) Effect of λ_3: Figure 4 (c) shows the effect of varying λ_3. Performance increases steadily up to 0.8, at which point the Rank-1 accuracy reaches 82.11% and the mAP reaches 79.89%. Compared with λ_1 and λ_2, the shared loss exhibits greater robustness, and larger values contribute positively to performance, highlighting the importance of modality-shared representations in cross-modal learning.

4) Effect of m: Figure 4 (d) presents the results for different numbers of tokens. Both Rank-1 accuracy and mAP improve as m increases from 2 to 8, with the best performance observed at 81.95% Rank-1 and 79.75% mAP. When the number of tokens exceeds 8, performance declines slightly, indicating that excessively long prompts may introduce redundancy or noise. Therefore, setting m to 8 provides an effective balance between semantic richness and optimization stability.

Overall, the hyperparameter analysis shows that a balanced weighting of the three losses is essential for optimal cross-modal feature learning, with $\lambda_1 = 0.7$, $\lambda_2 = 0.5$, $\lambda_3 = 0.8$, and $m = 8$ achieving the best performance. Notably, this configuration is applied uniformly across all three benchmark datasets, and setting all weights to 1 already yields competitive results (Rank-1: 81.93%, mAP: 79.62%), indicating that the proposed loss combination is inherently stable and not overly sensitive to parameter selection. These findings validate the design of SCL and highlight the effectiveness of jointly leveraging multiple semantic factors for cross-modal representation learning.

E. Computational Efficiency Analysis

We compare MDCN with the baseline and CSDN in terms of model complexity and inference efficiency. As shown in Table VII, MDCN introduces additional parameters due to the three-branch architecture in SCL, yet its inference speed remains competitive. MDCN contains 61.4M parameters, which is higher than the baseline (40.6M) and CSDN (45.6M). This increase is expected, since SCL restructures the final stage of the image encoder into three parallel branches for extracting specific, shared, and contextual features, each introducing independent pooling and fully connected layers that contribute to the additional parameter count. Despite this, MDCN achieves an inference time of 33s per image, matching CSDN and only marginally slower than the single-branch baseline. This efficiency is because only the modality-contextual branch is used during inference, keeping the forward pass lightweight. Notably, MDCN obtains substantially higher retrieval accuracy at the same inference speed as CSDN, and the 21M parameter increase over the baseline translates into consistent performance



(a) Initial Distribution (b) Baseline Distribution (c) MDCN Distribution (d) Modality Distribution

Fig. 6: t-SNE visualization of feature distributions on SYSU-MM01. (a) Initial features are scattered and heavily overlapped. (b) Baseline features form partial clusters but with noticeable inter-class confusion. (c) MDCN features exhibit compact clusters and clear separations, indicating improved discriminability. (d) Joint visualization of visual features and modality-related text features, illustrating the explicit semantic alignment in the multimodal space.

gains across all benchmarks. For real-world deployment, this trade-off is acceptable, as the improvement in retrieval accuracy outweighs a moderate increase in model size.

TABLE VII: Comparison of model complexity and inference efficiency.

Method	Parameter Count	Inference Time
Baseline	40.6M	32s
CSDN	45.6M	33s
MDCN	61.4M	33s

F. Visualization Analysis

1) Attention Maps Visualization: We employ Grad-CAM [64] visualization to highlight the attention regions activated by different feature types and prompt learning strategies, as shown in Figure 5. The original images are provided for reference. The heatmaps reveal that modality-specific features primarily capture modality-dependent details, modality-contextual features emphasize broader contextual semantics, and modality-shared features focus on cross-modal regions. When contextual features are refined through MRKD, the highlighted regions become more compact and semantically coherent, suggesting that MRKD suppresses noise and enhances semantic-aware discriminability. Regarding prompt learning, the standard strategy yields relatively scattered activations across body regions, with limited emphasis on key discriminative parts. In contrast, the proposed CPDL produces more concentrated responses in highly informative regions such as torso and limbs. These results indicate that CPDL not only improves cross-modal alignment but also strengthens the semantic focus of the learned representations.

2) Feature Distribution Visualization: To further evaluate the effectiveness of the proposed MDCN, we visualize the feature distributions using t-SNE [65], as illustrated in Figure 6. In the initial distribution (Figure 6 (a)), the features are highly scattered, showing weak cluster compactness and substantial overlaps across classes. After applying the baseline model (Figure 6 (b)), clusters begin to emerge. However, several issues remain: (i) features from the gray and purple classes overlap, making them indistinguishable; (ii) outlier black points are scattered far from the main clusters; and (iii) green cross-modality features remain dispersed instead of forming compact groups. In contrast, with MDCN (Figure 6 (c)), these issues are effectively mitigated. The features exhibit clearer intra-class compactness and stronger inter-class separability, demonstrating MDCN's ability to learn more discriminative and modality-robust representations. Furthermore, by jointly visualizing the text features of the modality-related prompts and the image features of both modalities (Figure 6 (d)), we observe that the textual features cluster together with the visual features of the corresponding semantics. This clearly demonstrates that our method has successfully learned meaningful modality information.

V. Discussion

A. Novelty

The novelty of the proposed Modality Decoupling and Coupling Network (MDCN) lies not merely in the incremental combination of vision-language models and feature separation, but in a fundamental paradigm shift regarding how identity and modality spaces interact. Unlike existing visible-infrared person re-identification (VI-ReID) frameworks that suffer from representation degradation, MDCN introduces three core conceptual and architectural breakthroughs:

Traditional learnable text prompts inevitably fall into a dilemma of passive entanglement, where modality prompts accidentally absorb identity cues and person prompts are forced to accommodate massive cross-modal visual discrepancies. CPDL breaks this limitation by pioneering an explicit semantic factorization mechanism. By decomposing a holistic modality-contextual prompt into independent, structurally isolated person-related and modality-related components, and enforcing a strict modality orthogonality constraint ($\mathcal{L}_{mo}$), we create a zero-redundancy boundary between identity definitions and sensor specificities. This shifts prompt learning from traditional empirical tuning to a mathematically sound, disentangled textual anchor system.

Existing feature-level alignment techniques usually rely on brute-force image-text matching or feature concatenation, which renders the model highly susceptible to real-world visual degradations such as low-light noise, background clutter, and severe illumination variations. SCL redefines cross-modal alignment by transforming these decoupled text anchors into frozen, high-level semantic class prototypes. Because natural language represents absolute, high-level abstract concepts immune to pixel-level noise, forcing a three-branch visual encoder to geometrically converge toward these stable textual prototypes establishes a domain-invariant coordinate system in the multimodal latent space. This conceptual leap regularizes chaotic visual feature distributions into structured, noise-resistant semantic clusters rather than passively matching local visual patterns.

While standard knowledge distillation methods focus on duplicating vanilla feature representations, they fail to inherit the underlying geometric topology of heterogeneous feature spaces. MRKD addresses this by introducing a hierarchical cosine angular distillation paradigm that operates on a philosophy of selective structural forgetting. By deliberately discarding cross-modal relations in the specific branch and intra-modal details in the shared branch, we proactively prevent cross-space semantic contamination. This tactical erasure forces the holistic modality-contextual space (f_{co}) to inherit only the pure relational essence—simultaneously gaining intra-modal compactness and cross-modal consistency. MDCN thus pioneers a structurally complementary, relationally enriched representation framework that eliminates semantic confusion during multi-space information fusion.

In summary, through the synergistic interplay of CPDL, SCL, and MRKD, MDCN establishes a semantically clean, structurally complementary, and relationally enriched cross-modal representation learning framework that significantly elevates cross-spectral identity retrieval.

B. Limitations

Although the proposed MDCN achieves strong cross-modal retrieval performance under standard real-world degradations, it is worth discussing its bound behavior under severe, structural partial occlusion notorious challenge in surveillance analytics. Partial occlusion introduces two deep-seated spatial difficulties for VI-ReID. First, when a pedestrian is significantly blocked by persistent physical obstacles (e.g., vehicles or pillars), these long-range occlusions do not merely introduce pixel-level noise, but induce acute semantic drifts during the visual-textual alignment phase. Since the continuous text tokens in Stage I are optimized via image-text contrastive constraints, prolonged structural missingness forces the learnable prompts to inadvertently assimilate the semantic attributes of the occluding entities rather than the actual identities. This localized contextual distortion ultimately perturbs the geometric topology of the hierarchical angular relations in MRKD. Second, severe occlusion fundamentally triggers severe spatial feature misalignment across non-overlapping views, where key discriminative parts present in the query modality are entirely absent in the gallery spectrum.

To mitigate these adverse effects, two future dimensions appear promising. First, a part-level feature extraction mechanism could be integrated into the multi-branch visual network to explicitly focus on visible, unobstructed body regions, thereby filtering out occlusion-induced noise at the pixel level. Second, a local matching strategy can be adopted during the inference phase, where fine-grained part-to-part correspondences are established to explicitly alleviate the cross-view cross-spectral feature mismatch problem. We leave these extensions for future exploration and believe that augmenting the MDCN paradigm with part-based structural reasoning will further solidify its robustness under complex, real-world deployment conditions.

VI. Conclusion

This paper develops a novel MDCN method to address the challenges of VI-ReID. To the best of our knowledge, MDCN is the first method to introduce modality-contextual text prompts. It jointly leverages CPDL and SCL to facilitate effective multi-modal, multi-feature space learning. Specifically, CPDL uses prompts to decouple features and enforce modality orthogonality, ensuring that modality information is clearly distinguished from identity information. SCL embeds the learned high-level semantic information into visual representations, guiding the image encoder to emphasize critical identity-related attributes. Furthermore, MRKD is proposed to integrate valuable high-level semantic relationships from both modality-shared and modality-specific feature spaces into the modality-contextual feature space, significantly enhancing the extraction of identity-aware valuable information. Extensive experiments on SYSU-MM01, RegDB, and LLCM datasets demonstrate that MDCN consistently achieves robust cross-modal feature alignment and surpasses existing methods under challenging visible-infrared settings. Visualization analyses further reveal that MDCN produces more compact and discriminative feature clusters, effectively emphasizing cross-modal semantic regions while reducing noise. Beyond accuracy gains, our method shows strong potential for real-world cross-spectral surveillance applications, where reliable identity matching under diverse lighting conditions is essential. The integration of CPDL, SCL, and MRKD not only enhances representation discriminability but also establishes a scalable pathway for broader multi-modal learning paradigms.

With the rapid convergence of vision and language technologies, extending MDCN to incorporate textual cues or human-in-the-loop feedback may further strengthen its applicability to intelligent monitoring, forensic analysis, and cross-modal security analytics. For example, when a target is partially occluded, operators could interactively adjust the learned person-related prompts, such as emphasizing a specific clothing color, to refine retrieval of ambiguous identities on the fly without retraining. Such interactive prompt tuning injects human prior knowledge into the frozen semantic anchors, making the system more adaptable to hard cases. We believe that jointly modeling visible, infrared, and auxiliary textual information, augmented with flexible human guidance, will pave the way for more accurate, interpretable, and trustworthy cross-modal retrieval in practical deployments.

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